of Altered Forest Ecosystems) project.

Scottish barnacles

In 1961, American ecologist Joseph Connell published the results of his research on barnacles on the Scottish coast. Since free-swimming barnacle larvae can settle anywhere, Connell had tested why the lower part of the intertidal zone was colonized by *Balanus balanoides* barnacles and the upper part by *Chthamalus stellatus*. He wanted to know if this was due to competition, predation, or environmental factors.

Connell manipulated the local environment, and monitored it for over a year. In one area, he removed the *Chthamalus* barnacles. They were not replaced by *Balanus*, which suggested that *Balanus* could not tolerate the desiccation that occurred in the upper zone at low tide. Connell then removed the *Balanus* population from the lower zone, and found that *Chthamalus* barnacles did replace them. Both species could live in the lower zone, but only one could survive higher up. This suggested that *Chthamalus* was better able to deal with the harsh conditions of the upper zone, but was outcompeted by *Balanus* lower down. The “fundamental niche” of *Chthamalus* (where the species would normally be able to survive) encompassed both zones, but its “realized niche” (the actual area it inhabits) was more restricted.